

Question 1 Correct Mark 1.00 out of 1.00 [Flag question](#)

Convert the following algorithm into a program and find its time complexity using the counter method.

void function (int n)

```
{
    int i= 1;

    int s =1;

    while(s <= n)
    {
        i++;
        s += i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main() {
4     int n, count = 0;
```

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main() {
4     int n, count = 0;
5     scanf("%d", &n);
6
7     int i = 1, s = 1;
8     int k = 0;
9
10    while (s <= n) {
11        i++;
12        s += i;
13        k++;
14    }
15
16    count = 3 * k + 3;
17
18    printf("%d\n", count);
19    return 0;
20 }
21
22
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.